

Mini Project – Guidelines

The mini project should demonstrate:

- ✓ Practical implementation of database concepts
- ✓ Database design using ER modeling
- ✓ Table creation with constraints
- ✓ CRUD operations (Create, Read, Update, Delete)
- ✓ Queries, joins, views
- ✓ Stored procedures / triggers
- ✓ Transaction management (ACID properties)
- ✓ Project Development Phases

Phase 1: Requirement Analysis

- ✓ Identify system purpose
- ✓ Define user roles
- ✓ List required functionalities
- ✓ Identify input and output

Phase 2: System Design

1. Identify Entities

- Customer
- Product
- Orders
- Supplier

2. Draw ER Diagram

3. Define Relationships

Phase 3: Database Implementation

- ✓ Create database
- ✓ Create tables
- ✓ Apply constraints
 - PRIMARY KEY
 - FOREIGN KEY
 - UNIQUE
 - NOT NULL
 - CHECK
- ✓ Insert sample data

Phase 4: Query Implementation

Must include:

- ✓ SELECT
- ✓ WHERE
- ✓ GROUP BY
- ✓ ORDER BY
- ✓ JOINS
- ✓ SUBQUERIES
- ✓ VIEWS

Phase 5: Advanced Concepts

At least 2 of the following:

- ✓ Stored Procedures
- ✓ Triggers
- ✓ Transactions
- ✓ Indexing
- ✓ Backup & Recovery
- ✓ Normalization (1NF, 2NF, 3NF)
- ✓ Role-Based Access

Phase 6: Frontend Integration

- ✓ HTML forms
- ✓ PHP / Python / Java backend
- ✓ Database connectivity

Project Report Format:

The report should include:

1. Title Page
2. Bonafide Certificate
3. Declaration
4. Acknowledgment
5. Abstract
6. Table of Contents

7. Introduction
8. Literature Review
9. System Analysis
10. ER Diagram
11. Database Schema
12. SQL Implementation
13. Testing
14. Screenshots
15. Conclusion
16. Future Scope
17. References

Sample: **PHARMACY MANAGEMENT SYSTEM**

Technologies Used

Frontend → HTML, CSS

Backend → PHP

Database → MySQL

Server → XAMPP / WAMP

Architecture Flow

User → HTML Form → PHP → MySQL Database → Response → Browser

STEP-BY-STEP PROCESS

1. Create Database

```
CREATE DATABASE PharmacyDB;  
USE PharmacyDB;
```

2. Create Tables

```
CREATE TABLE Supplier (
```

```
supplier_id INT PRIMARY KEY AUTO_INCREMENT,  
supplier_name VARCHAR(100),  
phone VARCHAR(15),  
address VARCHAR(200)  
);
```

```
CREATE TABLE Medicine (  
    medicine_id INT PRIMARY KEY AUTO_INCREMENT,  
    medicine_name VARCHAR(100),  
    price DECIMAL(8,2),  
    quantity INT,  
    expiry_date DATE,  
    supplier_id INT,  
    FOREIGN KEY (supplier_id) REFERENCES Supplier(supplier_id)  
);
```

3. Frontend design (HTML)

```
<!DOCTYPE html>  
<html>  
<head>  
    <title>Add Medicine</title>  
</head>  
<body>
```

```
<h2>Add Medicine</h2>
```

```
<form action="add_medicine.php" method="post">  
    Medicine Name: <input type="text" name="medicine_name"  
required><br><br>  
    Price: <input type="number" step="0.01" name="price"  
required><br><br>  
    Quantity: <input type="number" name="quantity" required><br><br>  
    Expiry Date: <input type="date" name="expiry_date" required><br><br>  
    Supplier ID: <input type="number" name="supplier_id"  
required><br><br>
```

```
        <input type="submit" value="Add Medicine">
    </form>
```

```
</body>
</html>
```

4. Database Connectivity (PHP)

```
<?php
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "PharmacyDB";

$conn = new mysqli($servername, $username, $password, $dbname);

if ($conn->connect_error) {
    die("Connection Failed: " . $conn->connect_error);
}
?>
```

5. Backend (Insert data)

```
<?php
include 'db_connect.php';

$medicine_name = $_POST['medicine_name'];
$price = $_POST['price'];
$quantity = $_POST['quantity'];
$expiry_date = $_POST['expiry_date'];
$supplier_id = $_POST['supplier_id'];

$sql = "INSERT INTO Medicine
```

```
(medicine_name, price, quantity, expiry_date, supplier_id)
VALUES ('$medicine_name', '$price', '$quantity', '$expiry_date',
'$supplier_id');
```

```
if ($conn->query($sql) === TRUE) {
    echo "Medicine Added Successfully!";
} else {
    echo "Error: " . $conn->error;
}
```

```
$conn->close();
?>
```

6. DISPLAY MEDICINES

```
<?php
include 'db_connect.php';
```

```
$sql = "SELECT * FROM Medicine";
$result = $conn->query($sql);
```

```
echo "<h2>Medicine List</h2>";
echo "<table border='1'>
<tr>
<th>ID</th>
<th>Name</th>
<th>Price</th>
<th>Quantity</th>
<th>Expiry</th>
</tr>";
```

```
while($row = $result->fetch_assoc()) {
    echo "<tr>
<td>".$row["medicine_id"]."</td>
<td>".$row["medicine_name"]."</td>
```

```

        <td>".$row["price"]."</td>
        <td>".$row["quantity"]."</td>
        <td>".$row["expiry_date"]."</td>
    </tr>";
}

```

```

echo "</table>";
$conn->close();
?>

```

7. SALES MODULE (Insert Sale)

```

CREATE TABLE Sales (
    sale_id INT PRIMARY KEY AUTO_INCREMENT,
    medicine_id INT,
    quantity_sold INT,
    total_price DECIMAL(10,2),
    sale_date DATE,
    FOREIGN KEY (medicine_id) REFERENCES Medicine(medicine_id)
);

```

```

<?php
include 'db_connect.php';

```

```

$medicine_id = $_POST['medicine_id'];
$qty = $_POST['quantity_sold'];

```

```

$result = $conn->query("SELECT price, quantity FROM Medicine WHERE
medicine_id=$medicine_id");
$row = $result->fetch_assoc();

```

```

if($row['quantity'] >= $qty){

```

```

    $total = $row['price'] * $qty;

```

```

$conn->query("INSERT INTO Sales(medicine_id, quantity_sold,
total_price, sale_date)
VALUES($medicine_id, $qty, $total, CURDATE())");

$conn->query("UPDATE Medicine SET quantity = quantity - $qty
WHERE medicine_id=$medicine_id");

echo "Sale Successful!";
} else {
echo "Insufficient Stock!";
}

$conn->close();
?>

```

FOLDER STRUCTURE

PharmacyProject/

```

|— add_medicine.html
|— add_medicine.php
|— sell_medicine.php
|— view_medicine.php
|— db_connect.php

```
